

ZACHARY RYAN LOPEZ

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SUMMARY

AI/ML Data Operations specialist with 7+ years of experience across Apple data operations, annotation QA, and workflow improvement. Builds Python and SQL tools, validates high-volume datasets, and improves labeling consistency, reporting speed, and escalation handling.

CORE SKILLS

Python, SQL, Pandas, SQLite, Tableau, Power BI, Jupyter, Git, GitHub, Jira, Confluence, Excel

Annotation QA, MLops, Data Operations, ETL pipelines, data validation, error analysis, taxonomy governance, PII handling, NLP evaluation, RLHF, prompt engineering, A/B testing, ontology design, UX evaluation

Cross-functional collaboration, stakeholder communication, SOP authoring, onboarding, playbooks, de-escalation, project coordination, workflow improvement

WORK EXPERIENCE

Independent — AI/ML Data Operations Researcher | Austin, TX | **Jun 2024–Present**

- Built a production-grade ML incident-response system in Python, FastAPI, and async SQLAlchemy with audit trails, SEV1–SEV3 state handling, versioned runbooks, and governance artifacts.
- Developed an annotation analytics dashboard tracking labeling quality trends, ML support metrics, and workflow performance; deployed at zrl.dev.
- Engineered ETL and data-validation pipelines with Pandas and SQLite, automating quality checks and reducing reporting time by 40%; portfolio site built and deployed via Next.js, Tailwind CSS, and Vercel.

Apple — Data Operations Analyst | Austin, TX | **Sep 2021–Jun 2024**

- Partnered with global cross-functional teams to identify and close annotation workflow gaps, reducing escalation resolution time by 43%.
- Audited high-volume audiovisual datasets against Siri NLP guidelines, maintaining policy-compliant labeling through fast-paced review cycles.
- Applied guideline-based QA and linguistic judgment to ambiguous utterances, improving labeling consistency across NLP annotation pipelines.

Apple — Technical Expert | Austin, TX | **Mar 2019–Sep 2021**

- Diagnosed recurring faults via log analysis and trend pattern review, reducing repeat contacts by 20%; authored playbooks adopted by 5+ teammates to improve first-time resolution rates and triage speed.
- Resolved 1,200+ escalations with a 95% first-time fix rate and 4.9/5 CSAT using targeted communication and structured de-escalation techniques.

Apple — Business Development Specialist (Apple Pay) | Austin, TX | **Jan 2021–Jun 2021**

- Prospected hundreds of e-commerce merchants across major platforms and supported Apple Pay onboarding, integration, and adoption efforts.
- Developed tailored value propositions based on API capabilities and partner needs, reducing adoption resistance through direct email and Webex collaboration.

Apple — AppleCare Support Advisor | Austin, TX | **Mar 2020–Jan 2021**

- Delivered 90%+ same-call resolution across 2,000+ cases through remote diagnostics and knowledge-base workflows, minimizing escalations.

Apple — Specialist | Austin, TX | **Apr 2017–Mar 2019**

- Drove \$10K+ in monthly sales with a 30% attach rate and helped 10+ SMBs adopt Apple ecosystem workflows.

Starbucks — Shift Lead | Austin, TX | **Sep 2016–Mar 2017**

- Led shifts with 7+ baristas, coordinated task delegation and break coverage, and mentored 4+ new hires.

Target — Guest Service Attendant / Front-End Supervisor | Austin, TX | **Aug 2015–Nov 2016**

- Supervised 5+ team members, trained 10+ team members; supported register flow and guest services.

SELECTED PROJECTS

ML Incident Response Playbook + Runbook System | May 2026–Present

- Built a production-grade ML incident-response system in Python, FastAPI, async SQLAlchemy, and structlog with audit trails and domain-enforced SEV1–SEV3 state handling.
- Authored 8 ML-specific runbooks and governance artifacts, including versioned model cards and change logs.
- Implemented security scanning with TruffleHog, Trivy, Cosign, and CodeQL.

Annotation Analytics Dashboard | Sep 2025–Present

- Built a real-time annotation analytics dashboard with live KPI updates, alert threshold detection, capacity forecasting, and six-tab ML workflow visibility; rebuilt from vanilla JS to Next.js, TypeScript, and Recharts; live at zrl.dev/projects/annotation-dashboard.

Frontend Web Development | Sep 2025–Present

- Deployed a production-hardened portfolio at zrl.dev built on Next.js 15, TypeScript, Tailwind CSS, and Framer Motion via Vercel; live annotation dashboard powered by Recharts; secured with Cloudflare DNS/proxy and Turnstile bot protection; includes Jest test suite, Vercel Speed Insights, and a live contact API.

Data Pipeline Prototypes | Mar 2025–Jan 2026

- Engineered ETL and data-validation pipelines with Pandas and SQLite, automating quality checks and generating Tableau-ready outputs, reducing reporting time by 40%.

EDUCATION

Concordia University Texas | B.A. Political Science, Psychology | Expected Jun 2027

Ennis High School | HS Diploma | Jun 2015

CERTIFICATES

Harvard University: **Programming with Python** (May 2026)

Kaggle by Google: **Intermediate Machine Learning, Pandas, Advanced SQL**, and **Python** (May 2026)

Google Cloud: **Introduction to Generative AI** (May 2026)

HubSpot Academy: **HubSpot SEO II** and **Service Hub Software** (May 2026)

Apple: **ACiT / Apple Certified iOS Technician** (Mar 2019)